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EE1030/EE2100: Matrix Theory

Lecture 18: Introduction to Matrices

Topics :

1. Introduction to Matrices
 2. Elementary Matrix Operations
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Matrices - Introduction and Notation

one way of storing the data $\rightarrow$ vectors $\vec{x}_1 = \begin{bmatrix} \end{bmatrix} \leftarrow$ Nutrition information of apple.
 $\downarrow$
 $\{\vec{x}_1, \dots, \vec{x}_m\} \in \mathbb{R}^n$
 $\downarrow$
 no. of attributes $\vec{x}_2 \rightarrow$ Nutrition facts of Banana.

Q1) a person consumes 2 apples & 3 banana's $\rightarrow$ Nutritional intake $\rightarrow 2\vec{x}_1 + 3\vec{x}_2$

Q2) Target is to achieve a particular Nutritional intake $\rightarrow \vec{b} \in \mathbb{R}^n$

a) can the target be achieved by consuming Apple & oranges? Yes if $\vec{b} \in \text{Span}(\vec{x}_1, \vec{x}_6)$
 $\alpha \vec{x}_1 + \beta \vec{x}_6 = \vec{b}$ where $\alpha, \beta \rightarrow$ no. of apples, no. of oranges
 $\alpha x_{1,1} + \beta x_{6,1} = b_1$
 $\vdots$
 $\alpha x_{1,n} + \beta x_{6,n} = b_n$
 Set of linear eq's $\leftarrow$

Matrix

Table 1: Nutrition facts of some fruits [source: FDA.GOV-link]

Fruit	Cal.	Fat [g]	Na [mg]	K [mg]	Carb. [g]	Fib. [g]	Sug. [g]	Pro. [g]	Ca %DV	Fe %DV	Vit. A %DV	Vit. C %DV
(1) Apple (242 g)	130	0	0	260	34	5	25	1	2%	2%	2%	8%
(2) Banana (110 g)	110	0	0	450	30	3	19	1	0%	2%	2%	15%
(3) Grapes (126 g)	90	0	15	240	23	1	20	0	2%	0%	0%	2%
(4) Kiwi (148 g)	90	1	0	450	20	4	13	1	4%	2%	2%	240%
(5) Lemon (58g)	15	0	0	75	5	2	2	0	2%	0%	0%	40%
(6) Orange (154 g)	80	0	0	250	19	3	14	1	6%	0%	2%	130%
(7) Pineapple (112 g)	50	0	10	120	13	1	10	1	2%	2%	2%	50%
(8) Strawberries (147 g)	50	0	0	170	11	2	8	1	2%	2%	0%	160%
(9) Watermelon (280 g)	80	0	0	270	21	1	20	1	2%	4%	30%	25%

Row and Column Vectors of a Matrix, Sub-matrices

$$A = \{\vec{a}_1, \dots, \vec{a}_m\} \in \mathbb{R}^n \rightarrow \text{Size of matrix?} = \begin{matrix} n \times m \\ \uparrow \\ \text{no. of rows} \end{matrix} \begin{matrix} \text{no. of columns.} \end{matrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \rightarrow \text{Consider it as collection of vectors} \Rightarrow \text{Each column of matrix is a vector}$$

$$\{\vec{a}_1, \dots, \vec{a}_n\} \in \mathbb{R}^m$$

$$\& \vec{a}_{i,j} = A_{ji}$$

$$A = [\vec{a}_1 \dots \vec{a}_n]$$

$$B = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ \vdots & \vdots & & \vdots \\ b_{m1} & \dots & & b_{mn} \end{bmatrix} \rightarrow \text{Consider entries of each row as a vector} \Rightarrow$$

matrix as a collection of row vectors

$$= \begin{bmatrix} \vec{c}_1^T \\ \vdots \\ \vec{c}_m^T \end{bmatrix} \text{ where } \vec{c}_i \in \mathbb{R}^n$$

$$\{\vec{b}_1, \dots, \vec{b}_m\} \in \mathbb{R}^{1 \times n}$$

$$i^{\text{th}} \text{ entry of the } i^{\text{th}} \text{ row vector } \vec{b}_i = b_{ij}$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} \vec{c}_1^T \\ \vec{c}_2^T \end{bmatrix} \text{ where } \vec{c}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\& \vec{c}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$= \{\vec{b}_1 = [1 \ 2 \ 3], \vec{b}_2 = [4 \ 5 \ 6]\}$$

Transpose $\uparrow$ (n)

$$(\vec{c}_1^T, \dots, \vec{c}_m^T) \text{ where } \vec{c}_i \in \mathbb{R}^n$$

$$i^{\text{th}} \text{ entry of } i^{\text{th}} \text{ vector } \vec{c}_i = b_{ij}$$

Some commonly encountered matrices

Matrix of Square if no. of rows = no. of columns

Matrix is Tall if $m > n$

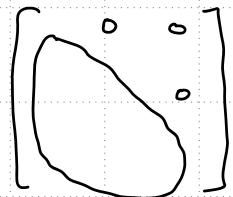
Matrix is Wide if $m < n$

Zero matrix $\mathbf{0} \in \mathbb{R}^{m \times n}$ = whose $\mathbf{0}_{ij} = 0 \quad \forall i, j = (1, \dots, m), (1, \dots, n)$

Identity Matrix : $\rightarrow$ Square & $(I_{ij}) = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$

Diagonal Matrix $\rightarrow$ " & $\mathbf{D}_{ij} = \begin{cases} d_i & \text{if } i=j \\ 0 & \text{otherwise} \end{cases}$

Square $\left\{ \begin{array}{l} \text{Lower triangular Matrix (L)} \Rightarrow (\mathbf{L})_{ij} = 0 \text{ when } j > i \\ \text{Upper triangular Matrix (U)} \Rightarrow (\mathbf{U})_{ij} = 0 \text{ when } j < i \end{array} \right.$



Transpose of a matrix $\Rightarrow$ if $\mathbf{B} = \mathbf{A}^T$ then $b_{ij} = a_{ji}$